

1a) It would be best to bid truthfully and rationally s.t.

$$v(x(p), p) - t(p) \geq v(x(\hat{p}), p) - t(\hat{p})$$

$$v(x(p), p) - t(p) \geq 0$$

1b) The contract has an expected cost of $E[\max(c_A, c_B)]$. We can use the CDF to find the actual cost. $F(x) = P(c_A \leq x) \cdot P(c_B \leq x)$.

Because each individual has equal probability to win the contract, and $x \in [100, 200]$, $F(x) = \left(\frac{x-100}{100}\right)^2$.

$$f(x) = 2 \left(\frac{x-100}{100} \right) \left(\frac{1}{100} \right) = 2 \left(\frac{x-100}{10000} \right).$$

$$\int_{100}^{200} x \cdot 2 \left(\frac{x-100}{10000} \right) dx \Rightarrow \frac{2}{10000} \int_{100}^{200} x(x-100) dx$$

$$\Rightarrow \frac{2}{10000} \int_{100}^{200} (x^2 - 100x) dx \Rightarrow \frac{2}{10000} \left(\frac{x^3}{3} - 50x^2 \right) \Big|_{100}^{200}$$

$$\Rightarrow \frac{2}{10000} \left(\frac{200^3}{3} - 50(200)^2 - \left(\frac{100^3}{3} - 50(100)^2 \right) \right) = \frac{2}{10000} (833,334)$$

= 166.67. So, Prof. Smith should expect to pay \$166.67.

For the RA's, $E[\min(c_A, c_B)]$. CDF $G(x) = 1 - P(\min(c_A, c_B) > x)$.
 $= 1 - P(c_A > x) \cdot P(c_B > x) = 1 - (1 - P(c_A \leq x))(1 - P(c_B \leq x))$. Probability is like before so $G(x) = 1 - \left(\frac{200-x}{10000}\right)^2$.

$$g(x) = 2 \left(\frac{200-x}{10000} \right) \Rightarrow \frac{2}{10000} \int_{100}^{200} x(200-x) dx = \frac{2}{10000} \int_{100}^{200} (200x - x^2) dx$$

$$\Rightarrow \frac{2}{10000} \left(100x^2 - \frac{x^3}{3} \right) \Big|_{100}^{200} \Rightarrow \frac{2}{10000} \left(100 \cdot (200)^2 - \frac{200^3}{3} - 100^3 + \frac{100^3}{3} \right)$$

$$\Rightarrow \frac{2}{10000} (666,666) = 133.33$$

$166.67 - 133.33 = 33.34$ exp. pay. Since each is equally likely to get the contract, each can expect \$16.67

1c) As found in (b), Anna gets 16.67 if selected.
 With investment her cost distribution becomes
 $c'_A \in [80, 180]$ due to effective cost lowering by 20.

When $c'_A < 100$, $P(c_B > c'_A) \Rightarrow A$ always wins = $\frac{1}{100}$
 When $c'_A \geq 100$, $P(c_B > c'_A) = \frac{200 - c'_A}{100}$

$$\begin{aligned}
 E[\text{payoff}] &= \int_{80}^{100} \int_{100}^{200} (c_B - c'_A) \cdot \frac{1}{10000} dc_B dc'_A \\
 &+ \int_{100}^{180} \int_{c'_A}^{200} (c_B - c'_A) \cdot \frac{1}{10000} c'_A dc_B dc'_A \\
 \text{1st:} \\
 \frac{1}{10000} \int_{80}^{100} \left(\frac{c_B^2}{2} - c'_A c_B \right) \Big|_{100}^{200} dc'_A &= \frac{1}{10000} \int_{80}^{100} [20000 - 200c'_A - 5000 + 100c'_A] dc'_A \\
 \Rightarrow \frac{1}{10000} [15000 - 100c'_A] \Big|_{80}^{100} &= \frac{1}{10000} [15000(100) - 50(100)^2 - 15000(80) \\
 + 50(80)^2] &= \frac{1}{10000} [1500000 - 500000 - 1200000 + 320000] = 12
 \end{aligned}$$

$$\begin{aligned}
 \text{2nd:} \\
 \frac{1}{10000} \int_{100}^{180} \left[\frac{c_B^2}{2} - c'_A c_B \right] \Big|_{c'_A}^{200} dc'_A &= \frac{1}{10000} \int_{100}^{180} [20000 - 200c'_A - \frac{c_B^2}{2} + c'_A c_B] dc'_A \\
 &= \frac{1}{10000} [20000c'_A - 100c'^2_A + \frac{c'^3_A}{6}] \Big|_{100}^{180} \\
 &= \frac{1}{10000} [20000(180) - 100(180)^2 + \frac{(180)^3}{6} - 20000(100) + 100(100)^2 - \frac{(100)^3}{6}] \\
 &= \frac{1}{10000} [165333.33] = 16.53
 \end{aligned}$$

$12 + 16.53 = 28.53 - 16.67 = 11.86 \therefore$ She should not pay more than \$11.86 on upfront investment

2a)

Allocation:

- giving both licenses to agent 2 presents the highest total value.

$$A+B \rightarrow 2 = 100$$

Transfer:

$$A \rightarrow 3 = 40$$

$$B \rightarrow 1 = 55$$

$\therefore$ w/o agent 2 cost is 95. This is total cost when not present since other agents pay nothing when 2 gets both.

$$u_2(0) = 100 - 95 = 5. \text{ Agent 2 gets a payoff of 5.}$$

2b)

A VCG mechanism explicitly does not allow for single agent diversion by DIC.

When shading down, they could lose a potentially positive payoff. Shading up leads to potentially overpaying, leading to disutility.

2c)

No, VCG is strategy proof. So, it doesn't make sense to coordinate with other agents.

3)

While the colleague is correct about what the revelation principle does, it is not optimal to only focus attention to mechanisms it applies to. There are equilibria which arise from indirect mechanisms, so restricting attention to direct mechanisms would screen out potential equilibria which may be what the professor desires.

4a)

Define:

$z \in \{0, 1\}$ be the indicator for if the public good is provided

$v_i \in [0, 1]$ be consumer valuations

Firm's Problem:

$$\max_{g(\cdot), t(\cdot)} E \left[\sum_{i=1}^N \underline{t_i(v_i)} - \underline{cN \cdot z(v_i)} \right]$$

s.t.

$$v_i \cdot E[z(v_i, v_{-i})] - E[t_i(v_i, v_{-i})] \geq v_i \cdot E[z(v'_i, v_{-i})] - E[t_i(v'_i, v_{-i})] \quad (IC)$$

$$v_i \cdot E[z(v_i, v_{-i})] - E[t_i(v_i, v_{-i})] \geq 0 \quad (IR)$$

4b)

$$Z_i(v_i) = E[v_i] (z(v_i, v_i))$$

$$T_i(v_i) = E[v_i] (t_i(v_i, v_i))$$

by envelope:

$$T_i(v_i) = v_i Z_i(v_i) - \int_0^{v_i} Z_i(s) ds - T_i(0)$$

$\Rightarrow$ by IR, $T_i(0) = 0$

$$T_i(v_i) = v_i Z_i(v_i) - \int_0^{v_i} Z_i(s) ds$$

plug into optimization:

$$\max_{z(\cdot)} E \left[\sum_{i=1}^N (v_i Z_i(v_i) - \int_0^{v_i} Z_i(s) ds) z(v_i) - cN z(v_i) \right]$$

Int by parts and uniform dist assumpt:

$$\max_{z(\cdot)} E \left[\sum_{i=1}^N \left(v_i - \frac{1-F(v_i)}{f(v_i)} \right) z(v_i) - cN z(v_i) \right]$$

$$\text{s.t. } F(v_i) = v_i, f(v_i) = 1$$

$$\max_{z(\cdot)} E \left[\sum_{i=1}^N (2v_i - 1) z(v_i) - cN z(v_i) \right]$$

set binary $z = 1$ when

$$\sum_{i=1}^N (2v_i - 1) > cN \Rightarrow \sum_{i=1}^N v_i > \frac{1+cN}{2}$$

4c) Given providing the good cost cN s.t. $c \in (0,1)$ with values $v_i \in [0,1]$. The firm's goal is to maximize revenue. In part (b) we found that

$$\sum_{i=1}^N v_i \geq \frac{1+cN}{2} \text{ or that the cost is incurred when the sum of valuations exceeds } \frac{1+cN}{2}.$$

To be DIC, truth-telling is a dominant strategy despite the reports of other agents. To be ex-post IR, no agent pays more than their v_i after all information is known.

Further, when $z(v_i) = 0$, $t_i(v_i) = 0$, or no one pays when the good is not provided.

This allows us to restrict attention to when $z(v_i) = 1$. In that case, we found that the cost is $\frac{1+cN}{2}$.

Let's assume each agent pays a fixed fee if the good is provided. If fee $\bar{c} > t_i(v_i)$ ex-post IR is violated. But, restricting attention to agents with $t_i(v_i) \geq \bar{c}$, we find a problem. Agents with v_i at the top of the distribution are then incentivized to report s.t. $t_i(\hat{v}_i) = \bar{c}$, a contradiction of DIC.